



Dashboard Data Transcript

Hello everyone, welcome to BITS PLANET DIGITAL. Let us continue our learning journey for the course Data Visualisation and Storytelling. Today we will be continuing our module Building Dashboards.

We will be now looking at specific elements of dashboard building. We will now look at what are the different types of dashboard data, discussion about performance dashboards and the common mistakes that are made while designing dashboards. You have seen the data types that are used while making visuals.

Similar types of data are also figured out in the dashboard. However, we need to understand one more layer of information or one more layer of complexity that is at play when we talk about dashboards. When we design a visual, we are talking about the data elements within that one visual.

We have to understand when we talk about dashboard, it is a collection of visuals and the conformity of different types of data type, different types of data across the visuals so that a harmonious data story can be built in the dashboard. What do I mean by this? The data types have to be compatible across visuals so that the interactivity features can be seamlessly integrated into the dashboard. If this is not taken care, you might lead to a place where you are not able to create.

You might go be in a situation where the comparisons that you try to get across visuals using interactivity might lead to wrong conclusions. So, always be very clear about the different types of data that you have in your data structure and how are they going to interact as different visuals on the dashboard. Now, let us look at the different types of data that are used in the dashboard.

First and foremost is categorical data. It is always important to remember that your dashboard design is aimed towards driving action from the data insights it provides. Typically in a business context, when you are talking about the actions either in the strategy dashboard, operations dashboard or an analysis dashboard, you are looking at a particular unit's performance or a region sale or for that matter, a particular processes characteristics.

All this would form a categorical variable indicating the either sales or the uptime, some features like this. The variables that represent the category are the categorical variables. We have seen department, region and product.

It can be home products department in a particular region, say in Hyderabad or particular product, let us say cement or paint. What is the sale? So, sales is a common feature, quantitative feature you are trying to track. But there are three levels of category that you are trying to look at.



What is the sale of paint in Hyderabad from a particular department called home products. Then you have the quantitative data. Typically sales, calls handled or for that matter, volume of traffic, revenue generated, profits, GDP growth.

All these are quantitative numbers which you are trying to portray. And then you have time series which looks at the change over a period of time. And for this, you have the type aggregation function.

This is what you have to spend some time to understand. We will see in the live demo or in the experience and learning how time as a scale can be used to bring in more insights into the dashboard. So, let me just give you a small example and move on.

For example, you have sales for a month. It is a monthly time scale. You are seeing the monthly sales over a year.

Then what you can do? You want to understand how the sales are happening on weekdays across all the 12 months. Then you can dwell into one step below what we call a drill down from months to days, see the patterns. And on the days, you can go up to hours to see where is the peak load is coming.

Let us say about website traffic, you want to understand daily traffic versus hourly traffic and we can optimise the traffic to the website by looking at the patterns at a minute time scale. And they can be aggregated back to look at the overall picture. So, again, we need to strategically use this aggregation and disaggregation through drill down to create the insights that we want.

And then you have the comparison. Are they always the standard? You have typical comparisons between the different type of products, different type of regions or the same product over a period of time. Or, for example, you have the same product across different regions.

These kind of interactivity comparisons can be made. Finally, you have status indicators to show whether the target has been reached. How much is the gap? Are there any exceptions? So, for example, there is a sudden drop down in the activity.

You can have a red flashing saying that activity has fallen below this threshold, which means there is a problem. Or activities over shooting saying that there is high flow of traffic into the website showing that the threshold has been breached. You can build in predetermined levels of activity falling below or going above that can cause a trigger and draw your attention.

So, by this different type of data that you have on the dashboard, you can use them in the visuals and bring out the insights. Now, the focus here is to get actionable insights.

As I was saying, the whole aim of a dashboard is to drive action from the insights that you provide.

And always remember the type of variables, the data types that you have in the data. Always remember the different types of data that you have on the data set and how they are going to interact through different visuals on the dashboard. So, for this, I have taken the example of a dashboard from our Ministry of External Affairs, Government of India.

So, this dashboard is very interesting and very interactive. Here, for example, I have selected citizen services and you can play around by selecting trade and commerce or diaspora engagement. It will show the relevant parameters and the source is given here.

So, what are the categorical data that you see here? First of all, citizen services, trade and commerce, diaspora engagement is one level of categorical variable on which you have different types of services that are rendered by the Ministry of External Affairs. While I click this citizen services, it shows the visas granted, passports issued, mortal remains brought back, pilgrims assisted, so on and so forth. It has about seven categories in which these services are provided.

And again, you have the quantitative elements like the number of visas issued, passports issued, clearly depicted with the use of infographics, small elements, clearly giving you a picture of what is this services related to and what is the number of the services rendered by it. And this is the dashboard that it is there in the website when you take the first look. This is one level of information that is provided to you.

Now, once you want to go deep onto this, I click on visas granted, I end up on a dashboard which is like this. In this, you have a time series showing the visa issuances over a period of time, both regular visa, e-visa and number of e-visa countries are increasing over a period of time. And the pattern again in COVID, you can see there is a clear dip and that has been rising.

And what is most interesting here is the comparison is through time series graph, but also through the clear text box that shows that there is 4% decrease in visas issued in 2023-2024 compared to the previous period. And it also has a status thing, same as a text box saying that three days is the average issuance time for e-visa. Now, if you remember, simple text is also a very powerful visualisation tool that we have discussed in our earlier classes.

Here, they have simply used a simple text with a small numerical element saying that it is 4% decreased with the comparative time period. All of this is coming as a text box on the visual, automated of course. So, you can clearly bring your attention of the audience to two or three topics.



The topic that you want to highlight is the number of visas that you have issued, whether it is higher or lower compared to last period and what is the average time taken. Of course, if somebody is interested in further details, they have the graphs to help them do that comparison. All the five data types that we have seen on the categories of dashboard data are used in this particular dashboard, clearly bringing out insights on how visas are issued by our country.

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